VENAFI ADAPTABLE CA

Integration Guide

Applicable Devices:
KMES Series 3



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[1] DOCUMENT INFORMATION

[1.1] DOCUMENT OVERVIEW

The purpose of this document is to provide information regarding the configuration of the Futurex KMES Series 3 for integration with Venafi Trust Protection Platform (TPP) through its Adaptable CA functionality.

This document focuses on steps specific to Trust Protection Platform integration and assumes basic setup of the KMES Series 3 has been performed. For additional information and setup instructions for the KMES Series 3, see the relevant user guide. This document also assumes basic setup of Venafi Trust Protection Platform has been performed.

[1.2] ABOUT VENAFI

Trust Protection Platform

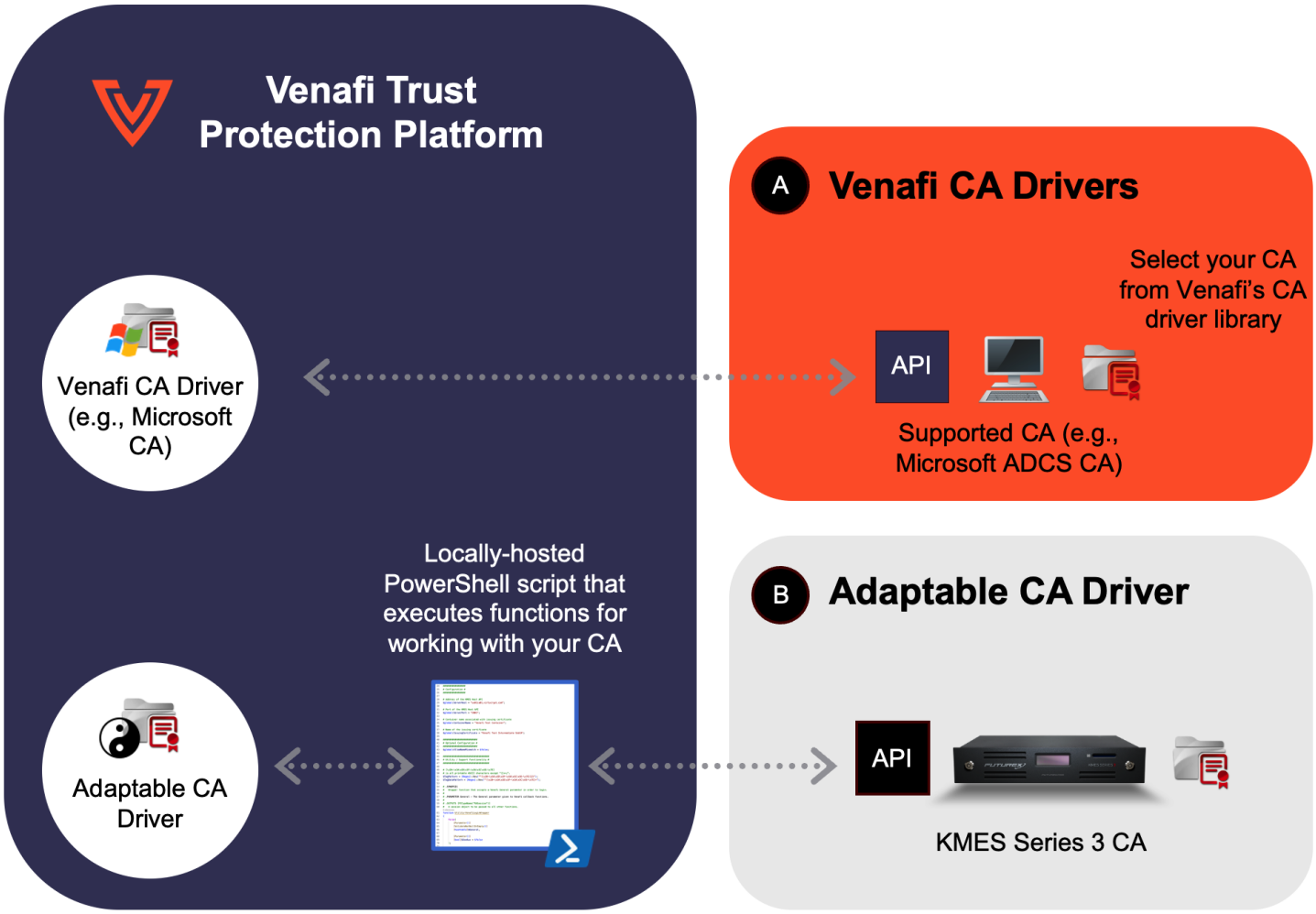
From Venafi's Trust Protection Platform datasheet: "Venafi Trust Protection Platform manages, secures and protects keys and certificates, delivering an enterprise-grade platform that provides enterprise-wide security, operational efficiency and organizational compliance."

Adaptable CA

From Venafi's documentation website: "Venafi builds and delivers software drivers designed to connect with many of the most common certificate authorities (CAs), application servers, and enterprise monitoring systems for notification use cases. But if the Venafi Driver Library does not include the driver you need, you can use a Venafi Adaptable Driver. In fact, because they are customizable, you can use an Adaptable driver in place of an existing driver to provide tighter integration between your business processes and Trust Protection Platform.

Adaptable drivers provide a common set of variables required by the majority of applicable use cases that are supported by Trust Protection Platform natively. Some Adaptable drivers also let you define additional text fields, yes/no (Boolean) fields, and a password credential field, which you can then use to elicit different behaviors or to pass additional data to the system or service to which you are integrating.

Adaptable drivers depend on a Microsoft PowerShell script hosted in your local environment to execute functions corresponding to standard certificate lifecycle stages or Trust Protection Platform events."


Venafi Trust Protection Platform
Venafi CA Driver (e.g., Microsoft CA)
Adaptable CA Driver
Locally-hosted PowerShell script that executes functions for working with your CA
A
Venafi CA Drivers
Select your CA from Venafi's CA driver library
API
Supported CA (e.g., Microsoft ADCS CA)
B
Adaptable CA Driver
API
KMES Series 3 CA

[2] PREREQUISITES

Before following the steps in this integration guide, the following prerequisite items must be installed and configured:

- Venafi Trust Protection Platform
- Futurex KMES Series 3 with minimum application version 6.3.1.3, the Registration Authority license enabled, and initial setup steps completed
- Futurex PowerShell scripts downloaded from the Venafi Marketplace

[3] KMES SERIES 3 CONFIGURATION

This section shows users the configurations they must make on the KMES Series 3 to allow Venafi to integrate through its Adaptable CA functionality.

[3.1] ENABLE THE REQUIRED HOST API COMMANDS

Since Venafi TPP will be connecting to the Host API port on the KMES, users must define which Host API commands will be enabled. To set the allowed commands, complete the following steps:

1. Log in to the KMES Series 3 application interface with the default Admin identities.
2. Navigate to **Administration**> **Configuration**>**Host API Options**, enable the commands listed below, then click **[Save]**.
    - **RAGX**: Retrieve Request (X.509 CSR)
    - **RASX**: Manipulate Signed Request
    - **RAUX**: Upload Request (X.509 CSR)
    - **RAYX**: Approve Requests
    - **RKLO**: Login User
    - **RKRK**: Retrieve Generated Keys

[3.2] CREATE KMES CREDENTIALS FOR VENAFI TPP

Venafi Trust Protection Platform supports two options for user credential management. The single user role option establishes one user per issuance policy that is permitted to submit certificate issuance requests, approve or deny requests, or revoke certificates. The dual user role option establishes two users per issuance policy, with one permitted only to submit certificate issuance requests and one permitted only to approve or deny issuance requests or revoke issued certificates.

For a greater degree of administrative separation as well as adherence to principles of role-based access control, Futurex recommends using the dual user method.

[3.2.1] Role and identity creation - Single user role option

The steps below outline how to set up a single user to control all aspects of certificate requesting, approval, and revocation.

1. Log in to the KMES Series 3 application interface with the default Admin identities.
2. Go to the **Identity Management** menu, select **Roles,** and click the **[Add...]** button. This will pull up the **Role Editor** dialog.
3. Specify a name for the role, set the number of logins required to **1**, and navigate to the **Advanced** tab and allow authentication to the **Host API** port only. All other fields can remain set with the default values.
4. Move to the **Permissions** tab and select the following permissions:

- **Certificate Authority** -> **Export**, **Upload**
- **Signing Approval** -> **Approve**

5. Click the **[OK]** button to finish creating the role.
6. Go to **Identities**, right-click anywhere on in the window and select **Add** > **Client Application**.
7. In the **Identity Editor** dialog:
    a. Under **Info**, select **Application** for the storage location, and specify a **name** for the identity.
    b. Under **Assigned Roles,** select the role you created.
    c. Under **Authentication**, remove the default API Key authentication mechanism, then add and configure the Password authentication mechanism.
    d. Leave all other fields as the default values and click the **[OK]** button to finish creating the identity.

[3.2.2] Role and identity creation - Dual user role option (Recommended)

The steps below outline how to set up a dual user role to control all aspects of certificate requesting, approval, and revocation. One user is solely responsible for submitting certificate issuance requests, and the other user is soley responsible for approving, rejecting, and revoking certificates.

1. Log in to the KMES Series 3 application interface with the default Admin identities.
2. Go to the **Identity Management** menu, select **Roles,** and click the **[Add...]** button. This will pull up the **Role Editor** dialog.
3. Specify a name for the role, set the number of logins required to **1**, and navigate to the **Advanced** tab and allow authentication to the **Host API** port only. All other fields can remain set with the default values.
4. Move to the **Permissions** tab and select the following permissions:
    - **Certificate Authority** -> **Export**, **Upload**
5. Click the **[OK]** button to finish creating the role.
6. Go to **Identities**, right-click anywhere on in the window and select **Add** > **Client Application**.
7. In the **Identity Editor** dialog:
    a. Under **Info**, select **Application** for the storage location, and specify a **name** for the identity.
    b. Under **Assigned Roles,** select the role you created.
    c. Under **Authentication**, remove the default API Key authentication mechanism, then add and configure the Password authentication mechanism.
    d. Leave all other fields as the default values and click the **[OK]** button to finish creating the identity.

Next, configure a second role and identity using the same steps outlined above, but enabling the **Approve** permission under the **Signing Approval** permission category instead.

[3.3] CREATE SIGNING APPROVAL GROUP

To enable certificate signing approval workflows on the KMES Series 3, a signing approval group must be created using the following steps:

1. Log in to the KMES Series 3 application interface with the default Admin identities.
2. Navigate to the **PKI** > **Signing Workflow** menu.
3. Select **[Add Approval Group...]**.
4. Specify a name for the approval group and click **[OK]** to save.
5. Right-click the newly-created approval group and select **Permission**.
6. Grant the **Venafi Adaptable CA** role the **Use** permission, then click **[OK]** to save.

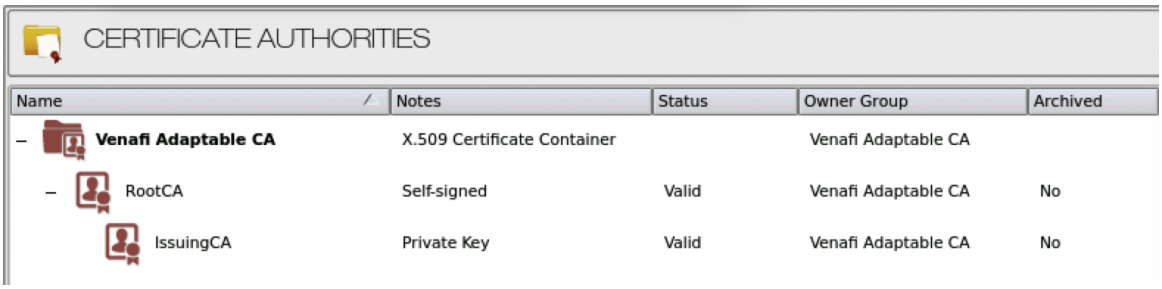
Note: Remember the group name because it will need to be referenced in the Adaptable CA driver configuration detailed later in this document.

[3.4] CREATE AN ISSUING CA

The steps below outline how to create an issuing CA certificate tree on the KMES Series 3.

1. Log in to the KMES Series 3 application interface with the default Admin identities.
2. Navigate to the **PKI** > **Certificate Authorities** menu.
3. Click the **[Add CA...]** button at the bottom of the page. This will open the **Certificate Authority** dialog.
4. Enter a name for the Certificate Container, select the Venafi Adaptable CA role in the Owner group drop-down menu, then click **[OK]**.
5. Right-click on the Certificate Container and select **Add Certificate** > **New Certificate...**
6. In the **Subject DN** tab, set a Common Name for the certificate, such as "RootCA".
7. In the **Basic Info** tab, leave the default values set.
8. In the **V3 Extensions** tab, select the **Certificate Authority** profile, then click **[OK]** to create the certificate.
9. Now, right-click on the root CA certificate and select **Add Certificate** > **New Certificate...**
10. In the **Subject DN** tab, set a Common Name for the certificate, such as "IssuingCA".
11. In the **Basic Info** tab, leave the default values set.
12. In the **V3 Extensions** tab, select the **Certificate Authority** profile, then click **[OK]** to create the certificate.

13. The root and issuing CA certificates are listed now inside the **Venafi Adaptable CA** Certificate Container.



[3.5] ADD AN ISSUANCE POLICY

1. Log in to the KMES Series 3 application interface with the default Admin identities.
2. Navigate to the **PKI** > **Certificate Authorities** menu.
3. Right-click on the issuing CA certificate and select **Issuance Policy** > **Add**, which will pull up the Issuance Policy dialog.
4. In the **Basic Info** tab:
    - Set an alias for the issuance policy (optional)
    - Set the desired number of approvals required for certificate requests
    - Set the hashes that you want to allow
5. In the X.509 tab:
    - Select the **Allow CSR uploads** checkbox
    - Set the **Default approval group** to the approval group you created in section 3.3
    - If you set the number of required approvals to 0 you must select the **Allow self approval** checkbox. Otherwise, leave it unchecked.
    - Add at least one **Extension Profile**
6. Click **[OK]** to save the issuance policy.

Note: If you set the number of approvals required to **0**, you must set Anonymous Signing security usage on the issuing CA certificate. To do so, right-click on the issuing CA certificate and select **Change Security Usage**. Then, in the drop-down menu, select **Anonymous Signing** and click **[OK]** to save the changes.

[3.6] ALLOW USER-DEFINED EXTENSIONS FOR X.509 EXTENSION PROFILES (OPTIONAL)

If you want to allow Subject Alternate Names (SANs) or other custom X.509 v3 extensions to be defined in Venafi Trust Protection Platform when creating certificates, you have two options.

[3.6.1] Enable the "Allow User-Defined Extensions" option for the X.509 Extension Profile

Follow the steps below to enable the "Allow User-Defined Extensions" option for the X.509 v3 Extension Profiles you plan to use with Venafi Adaptable CA:

1. Log in to the KMES Series 3 application interface with the default Admin identities.
2. Navigate to the **PKI** > **X.509 Extensions** menu.
3. Right-click the X.509 v3 Extension Profile you want to modify and select **Edit...**
4. Select the **Allow User-Defined Extensions** checkbox.
5. Click **[OK]** to save changes.

[3.6.2] Add Subject Alternate Names (SANs) to the X.509 Extension Profile

Follow the steps below to add Subject Alternate Names (SANs) to the X.509 v3 Extension Profiles you plan to use with Venafi Adaptable CA:

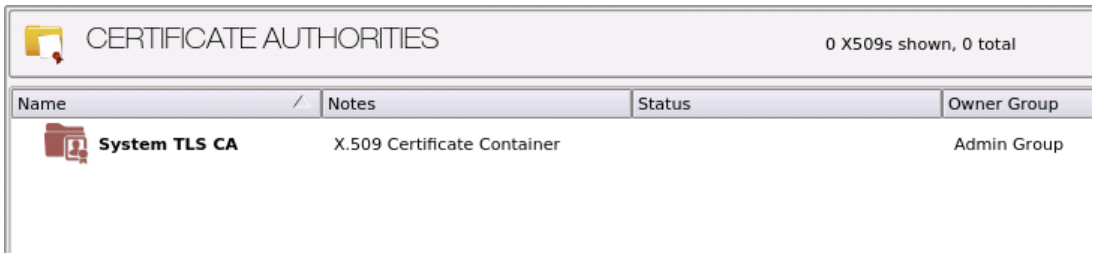
1. Log in to the KMES Series 3 application interface with the default Admin identities.
2. Navigate to the **PKI** > **X.509 Extensions** menu.
3. Right-click the X.509 v3 Extension Profile you want to modify and select **Edit...**
4. Click **[Add]** then select the **Subject Alternate Name** extension type in the drop-down menu and click **[OK]**.
5. Add at least one subject alternate names entry and click **[OK]**.
6. Click **[OK]** to save changes.

[4] CONFIGURING TLS CERTIFICATES FOR COMMUNICATION BETWEEN THE VENAFI TPP INSTANCE AND KMES SERIES 3

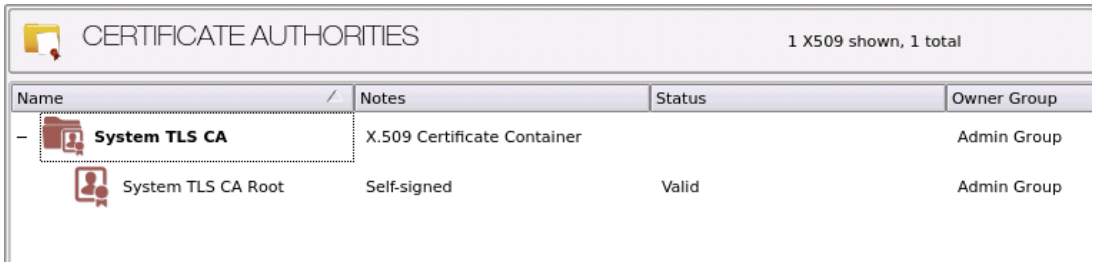
This section shows users how to set up TLS certificates for the connection between Venafi TPP and the System/Host API connection pair on the KMES Series 3.

[4.1] CREATE A CERTIFICATE AUTHORITY (CA)

1. Log in to the KMES Series 3 application interface with the default Admin identities.
2. Navigate to the **PKI** > **Certificate Authorities** menu.
3. Click the **[Add CA...]** button at the bottom of the page. This will open the **Certificate Authority** dialog.
4. Enter a name for the Certificate Container, select the Venafi Adaptable CA role in the Owner group drop-down menu, then click **[OK]**.
5. The Certificate Container that was just created will be listed now in the Certificate Authorities menu.

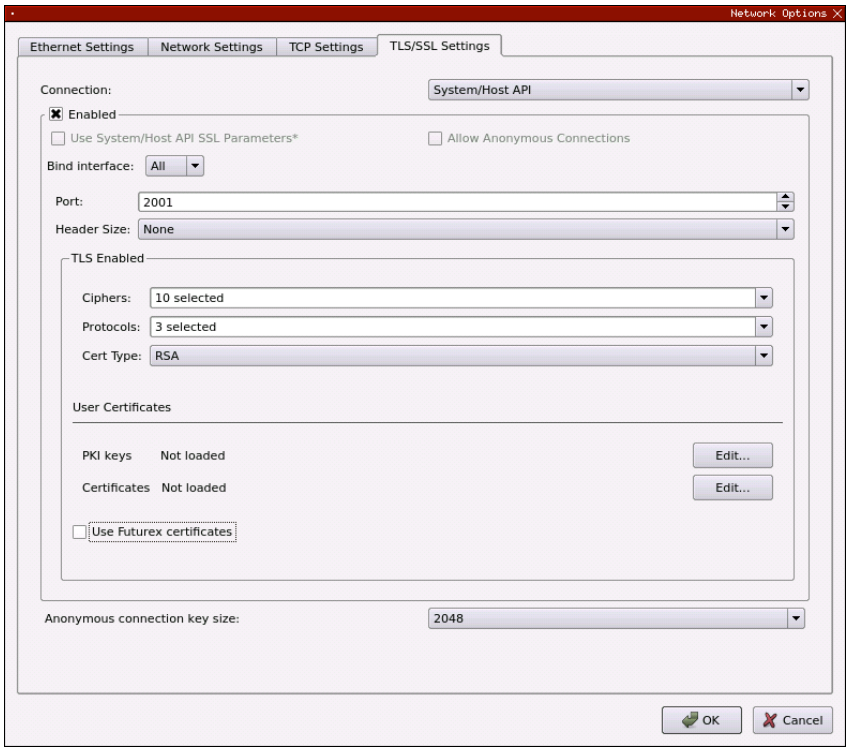


6. Right-click on the Certificate Container and select **Add Certificate** > **New Certificate...**
7. In the **Subject DN** tab, set a Common Name for the certificate, such as "System TLS CA Root".
8. In the **Basic Info** tab, leave the default values set.
9. In the **V3 Extensions** tab, select the **Certificate Authority** profile, then click **[OK]**.
10. The root CA certificate will be listed now under the previously created Certificate Container.



[4.2] GENERATE A CSR FOR THE SYSTEM/HOST API CONNECTION PAIR

1. Navigate to **Administration** > **Configuration** > **Network Options**.
2. In the **Network Options** dialog, select the **TLS/SSL Settings** tab.
3. Under the **System/Host API** connection pair, uncheck **Use Futurex certificates**, then click **[Edit...]** next to PKI keys in the User Certificates section.

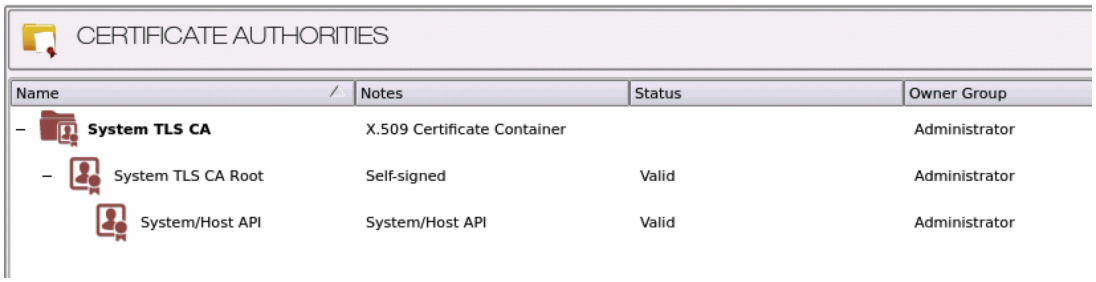


4. In the **Application Public Keys** dialog, click **[Generate...]**.
5. There will be a warning stating that SSL will not be functional until new certificates are imported. Select **[Yes]** if you wish to continue.
6. In the **PKI Parameters** dialog, set the encrypting key to **PMK**, key type to **RSA**, and key size to **2048**. Click **[OK]** to save.
7. It should show that a PKI Key Pair is loaded now in the **Application Public Keys** dialog. If this is the case, click **[Request...]**
8. In the **Subject DN** tab, you can leave the Common Name set to the default value of "System/Host API".
9. In the **V3 Extensions** tab, select the **TLS Server Certificate** profile.
10. In the **PKCS #10 Info** tab, select a save location for the CSR, then click **[OK]**.
11. There should be a message stating that the certificate signing request was successfully written to the file location that was selected. Click **[OK]**.
12. Click **[OK]** again to save the Application Public Keys settings.

13. In the main **Network Options** dialog, it should now show **Loaded** next to **PKI keys** for the System/Host API connection pair. Click **[OK]** to save and close the Network Options settings.

[4.3] SIGN THE SYSTEM/HOST API CSR

1. Navigate to the **PKI** > **Certificate Authorities** menu.
2. Right-click on the root CA certificate created in section 4.1, then select **Add Certificate** > **From Request...**
3. In the file browser, find and select the CSR that was generated for the System/Host API connection pair.
4. Once loaded, none of the settings need to be modified for the certificate. Click **[OK]**.
5. The signed System/Host API certificate should now show under the root CA certificate on the **Certificate Authorities** page.



[4.4] EXPORT THE ROOT CA CERTIFICATE

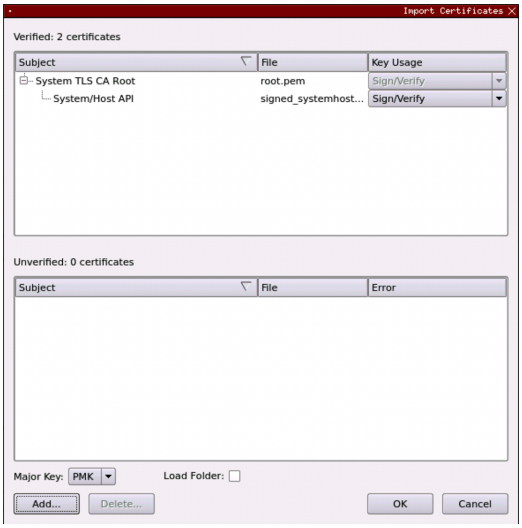
1. Navigate to the **PKI** > **Certificate Authorities** menu.
2. Right-click on the **System TLS CA Root** certificate, then select **Export** > **Certificate(s)...**
3. In the **Export Certificate** dialog, change the encoding to **PEM**, then click **[Browse...]**.
4. In the file browser, navigate to the location where you want to save the Root CA certificate. Specify "root.pem" as the name for the file, then click **[Open]**.
5. Click **[OK]**. A message box will pop up stating that the PEM file was successfully written to the location that you specified.

[4.5] EXPORT THE SIGNED SYSTEM/HOST API CERTIFICATE

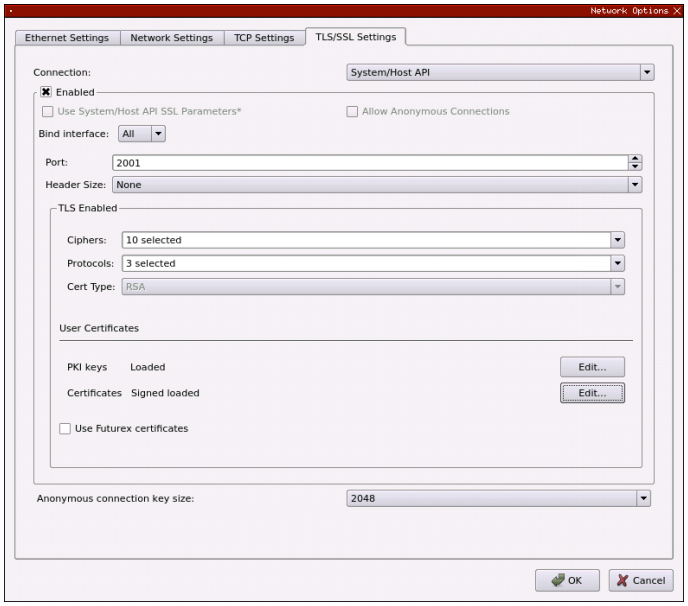
1. Go to **PKI** > **Certificate Authorities** menu.
2. Right-click on the **KMES** certificate, then select **Export** > **Certificate(s)...**
3. In the **Export Certificate** dialog, change the encoding to **PEM**, then click **[Browse...]**.
4. In the file browser, navigate to the location where you want to save the signed System/Host API certificate. Specify "signed_systemhostapi.pem" as the name for the file, then click **[Open]**.
5. Click **[OK]**. A message box will pop up stating that the PEM file was successfully written to the location that you specified.

[4.6] LOAD THE EXPORTED CERTIFICATES INTO THE SYSTEM/HOST API CONNECTION PAIR

1. Go to **Administration** > **Configuration** > **Network Options**.
2. In the **Network Options** dialog, select the **TLS/SSL Settings** tab.
3. Click **[Edit...]** next to Certificates in the User Certificates section.
4. Right-click on the **System/Host API SSL CA** X.509 Certificate Container, then select **[Import...]**.
5. Click **[Add...]** at the bottom of the **Import Certificates** dialog.
6. In the file browser, find and select both the root CA certificate and the signed System/Host API certificate, then click **[Open]**. The certificate chain should appear as shown below:



7. Click **[OK]** to save the changes. In the **Network Options** dialog, the System/Host API connection pair should show **Signed loaded** next to Certificates in the **User Certificates** section, as shown below:



8. Click **[OK]** to save and exit the Network Options dialog.

[4.7] ISSUE A CLIENT CERTIFICATE FOR VENAFI TPP

1. Go to **PKI** > **Certificate Authorities** menu.
2. Right-click on the **System TLS CA Root** certificate and select **Add Certificate** > **New Certificate...**
3. In the **Subject DN** tab, set a Common Name for the certificate, such as "Venafi".
4. All settings in the **Basic Info** tab should be left as the default values.
5. In the **V3 Extensions** tab, select the **TLS Client Certificate** profile, then click **[OK]**.
6. The Venafi certificate will be listed now under the **System TLS CA Root** certificate.

[4.8] EXPORT THE VENAFI CERTIFICATE AS PKCS #12 FILE

Note: To be able to perform the steps below you must go to **Configuration** > **Options** and enable the **Allow export of certificates using passwords** option.

1. Go to **PKI** > **Certificate Authorities** menu.
2. Right-click on the Venafi certificate, then select **Export** > **PKCS12...**
3. Make sure that the **Export Selected** option is selected, specify a unique name for the export file, then click **[Next]**.
4. Input a file password of your choosing, then click **[Next]**.
5. Click **[Finish]** to initiate the export.

Note: The **Venafi** certificate and the Root CA certificate that was exported in section 4.4 both need to be moved to the computer that will be running the Venafi TPP instance. In a later section, they will be configured and used for TLS communication with the KMES Series 3.

[5] CONFIGURING THE ADAPTABLE CA DRIVER

Two PowerShell scripts are required for connecting Venafi Adaptable CA interface to the KMES Series 3:

- Futurex KMES CA.ps1
- FuturexCreateCustomFields.ps1

Futurex KMES CA.ps1 must be copied into the AdaptableCA scripts folder, which is typically found at **C:\Program Files\Venafi\Scripts\AdaptableCA**.

FuturexCreateCustomFields.ps1 may be run from anywhere, as long as it is able to connect to Trust Protection Platform's web SDK. As Trust Protection Platform will attempt to enumerate the script as if it's a driver, Futurex recommends NOT placing it in the AdaptableCA scripts folder.

[5.1] CONFIGURING THE KMES SERIES 3 POWERSHELL DRIVER

To connect to a KMES Series 3, Trust Protection Platform uses a PowerShell configuration file containing user-defined parameters. This file can be opened in a text editor and contains the following fields:

```
##################
# Configuration #
##################

# Address of the KMES Host API
$global:ServerHost = "<KMES SERIES 3 HOSTNAME OR IP>";

# Port of the KMES Host API
$global:ServerPort = "<KMES SERIES 3 HOST API PORT>";

# Container name associated with issuing certificate
$global:ContainerName = "<CONTAINER NAME ON THE KMES SERIES 3>";

# Name of the issuing certificate
$global:IssuingCertificate = "<ISSUING CERTIFICATE ON THE KMES SERIES 3>";
```

You must install the System TLS CA Root certficate into the **Trusted Root Certificate Authorities** Store in Windows (otherwise certificate validation will fail).

To use multiple KMES Series 3 servers or even different issuing certificates, the user will need to create additional copies of the driver script with different filenames and manually change the variables for each instance required. If the driver script's filename is changed, it will appear in Trust Protection Platform with a different name.

[5.2] CONFIGURING THE CUSTOM FIELDS POWERSHELL SCRIPT

The second PowerShell file, FuturexCreateCustomFields.ps1, is a script that defines three custom fields in Trust Protection Platform. These are used for defining the approval group within the KMES Series 3 that will control approvals of issuance requests, as well as defining X.509 extension profiles. X.509 extension profiles allow users to define the type of certificate being deployed. This must match an option defined for the relevant issuance policy.

These fields are optional and can provide additional levels of granular control over Venafi policies for certificate attributes and issuance structure. These three fields are currently the only ones supported by Futurex and Venafi. These custom fields must be added using this script and cannot be manually added inside the Venafi application itself.

To configure the script, open FuturexCreateCustomFields.ps1 in a text editor and change the following variables to ones appropriate for the Venafi Trust Protection Platform installation.

```
# Configuration
$SdkUri = "<VENAFI TRUST PROTECTION PLATFORM HOSTNAME OR IP>"
$SdkUser = "<VENAFI TRUST PROTECTION PLATFORM USERNAME>"
$SdkPass = "<VENAFI TRUST PROTECTION PLATFORM PASSWORD>"
```

Once these changes have been made, the script can be run in PowerShell. This script only needs to be run once on each server running Venafi, regardless of how many KMES Series 3 units or issuance policies are defined.

[6] CONFIGURING VENAFI TPP TO USE THE FUTUREX ADAPTABLE CA DRIVER

This section outlines the configurations users must make in Venafi Trust Protection Platform before certificates can be requested through the Futurex Adaptable CA driver.

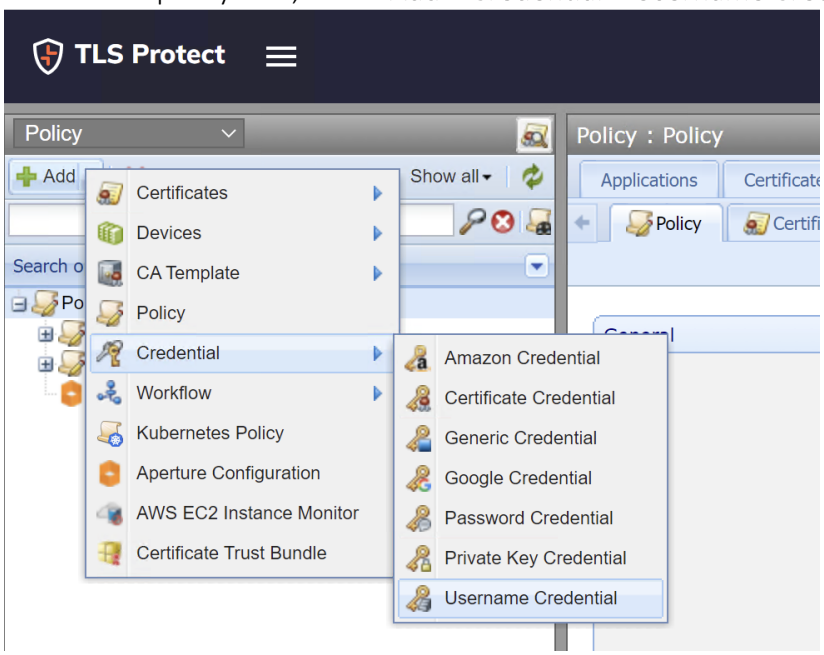
[6.1] CREDENTIAL MANAGEMENT

The identity and TLS client certificate that were created on the KMES Series 3 must be added as "credentials" in Venafi TPP.

[6.1.1] DEFINE USER CREDENTIALS

To define user credentials, perform the following steps:

1. Log in to Venafi Trust Protection Platform.
2. Select **Policy Tree** in the main menu.
3. In the main policy tree, select **Add** > **Credential** > **Username Credential**.



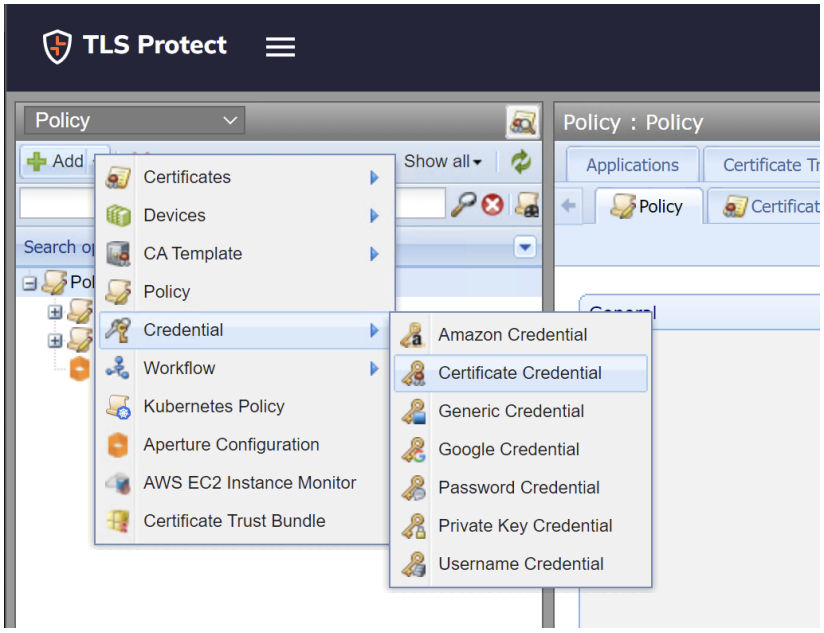
4. In the **Username Credential** window, add the username and password created for the Venafi identity on the KMES Series 3 earlier in the integration process, along with any other settings needed for the environment, such as a credential expiration date.
5. Click **Save** to save the credential.
6. Repeat steps 3-5 for each additional user needed.

[6.1.2] DEFINE TLS CLIENT CERTIFICATE CREDENTIALS

TLS client certificates are used to mutually authenticate with the KMES Series 3, allowing only authorized operation and establishing an encrypted tunnel to prevent man-in-the-middle eavesdropping on traffic.

To define TLS client certificate credentials, perform the following steps:

1. Log in to Venafi Trust Protection Platform.
2. Select **Policy Tree** in the main menu.
3. In the main policy tree, select **Add** > **Credential** > **Certificate Credential**.



4. In the **Certificate Credential** window, give the credential a name and choose the option to import a certificate and select the binary-encoded PFX/PKCS #12 certificate that you exported from the KMES Series 3 earlier in this integration guide.
5. Specify the corresponding private key password and begin the import process.
6. Once the certificate has been imported, select the **[Save]** button to complete the process.

[6.2] CA TEMPLATE CREATION

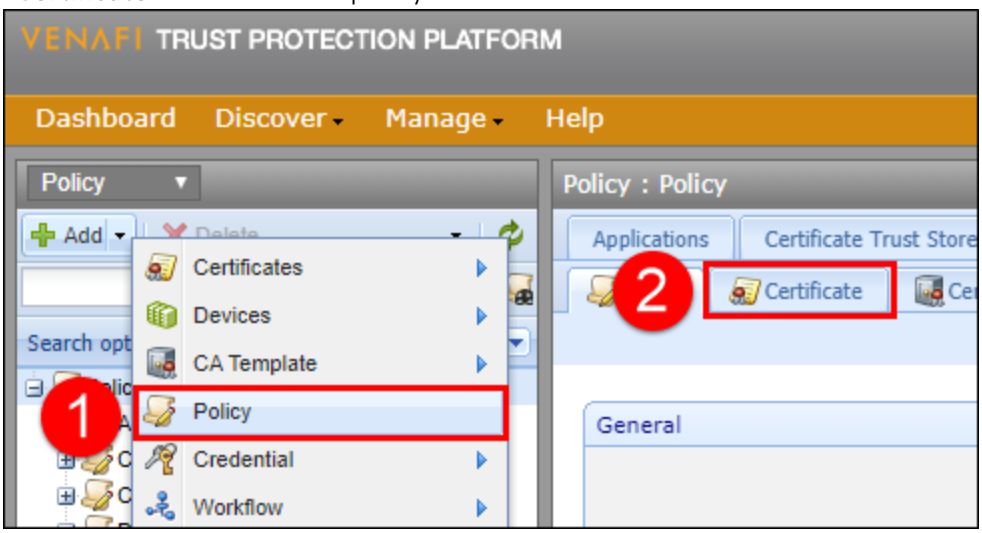
To create CA templates, perform the following steps:

1. Log in to Venafi Trust Protection Platform.
2. Select **Policy Tree** in the main menu.
3. In the main policy tree, select **Add** -> **CA Template** -> **Adaptable**. The **Add New Adaptable** window will appear.
4. Define the following General and Connection fields:
   a. **CA Name**: the desired CA name.
   b. **Username Credential**: the username credential created in section 6.1.1.
   c. **Certificate Credential**: the certificate credential created in section 6.1.2.
   d. **PowerShell Script**: Futurex KMES CA
5. If custom X.509 extensions or Futurex approval groups are desired, define them in the Custom Fields section. Note that for these to be visible, the custom fields PowerShell script defined earlier in this document must have been successfully run.
6. Select **Validate** to test the connection and authentication with the KMES Series 3. This can take 5-15 seconds to complete.
7. Select the **[Save]** button to complete the process.

[6.3] CERTIFICATE POLICY CREATION

To create certificate policies, perform the following steps:

1. Log in to Venafi Trust Protection Platform.
2. Select **Policy Tree** in the main menu.
3. In the main policy tree, select **Add** > **Policy**. The **Add New Policy** window will appear.
4. Define the policy name and any other desired settings and select **[Save]**.
5. Navigate to the **Certificate** tab of the new policy.



6. In the **Other Information** section, click the three dots next to the **CA Template** field and select the CA Template created in section 6.2.
7. Select the **[Save]** button to complete the process.

[7] TESTING A CERTIFICATE REQUEST, APPROVAL, AND ISSUANCE

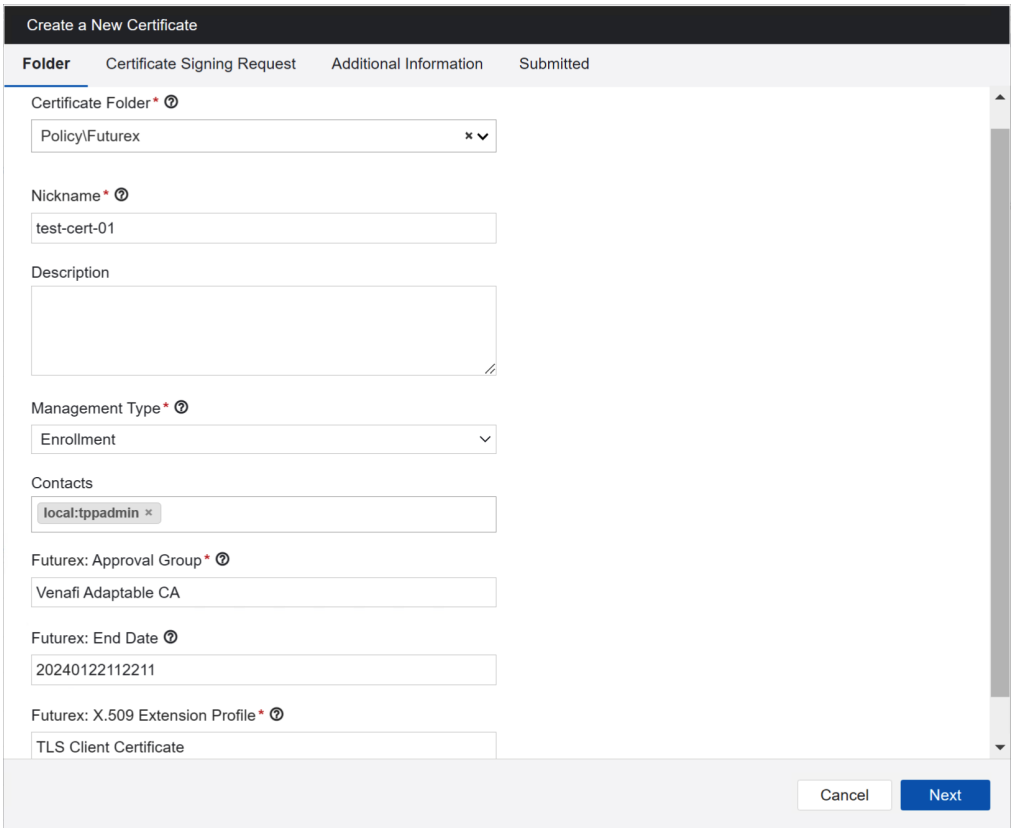
This section outlines the process for initiating a certificate request in Venafi TPP through the Futurex Adaptable CA driver, approving the request on the KMES, and viewing the issued certificate in Venafi TPP.

[7.1] INITIATE A CERTIFICATE REQUEST IN VENAFI TPP

To initiate a certificate request in Venafi TPP, perform the following steps:

1. Log in to Venafi Trust Protection Platform.
2. Select **Inventory** > **Certificates** in the main menu.
3. Select the **[Create a New Certificate]** button in the upper-right-hand corner of the page.
4. In the **Certificate Folder** drop-down menu, select the certificate policy you created in section 6.3, then specify the required values in the fields that populate. Click **[Next]** when finished.

   Example:

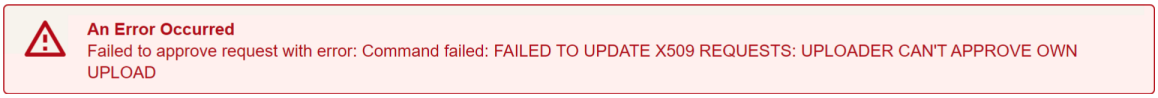


   Note: You must select **Enrollment** for the Management Type.

   Note: The End Date of the certificate must be specified in the format YYYYMMDDhhmmss.

5. In the CSR Generation drop-down menu select **Generate a CSR for me**, specify all of the desired values for the Certificate Signing Request, then click **[Next]** when finished.
6. Specify any additional information you wish to include in the request, such as Subject Alternate Names (DNS). Then, click **[Create Certificate]**. You should see a message stating the the certificate request has been submitted.

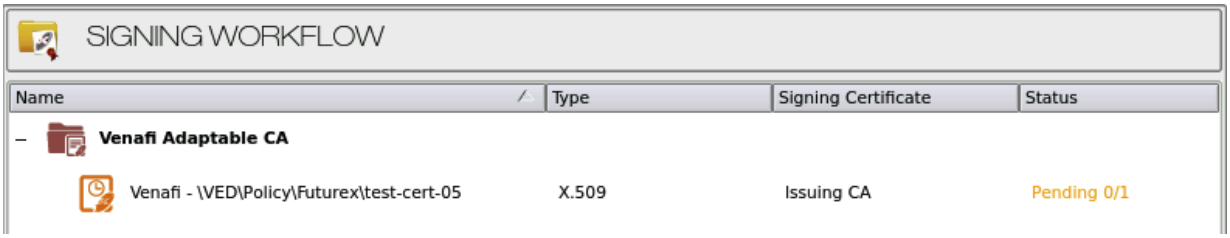
Note: If you configured the dual user roles option in section 3.2.2 to separate certificate requesting and certificate approval responsibilities, you may see the following message after submitting the certificate request. This is the expected behavior.



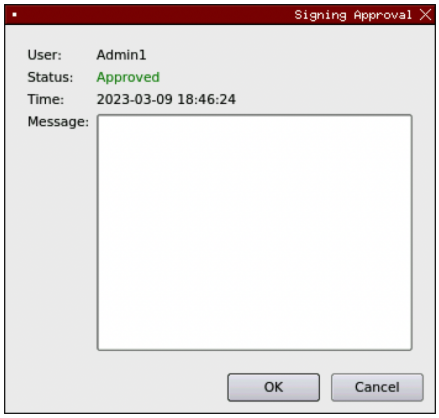
[7.2] APPROVE THE CERTIFICATE REQUEST ON THE KMES

To approve the certificate request on the KMES, perform the following steps:

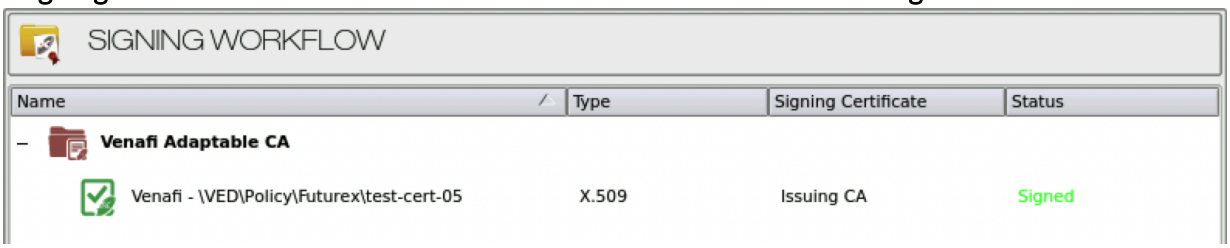
1. Log in to the KMES Series 3 application interface with the default Admin identities.
2. Navigate to **Administration**> **Signing Workflow** menu. You should see the certificate signing request inside your Venafi Adaptable CA approval group.



3. Right-click on the pending certificate request and select **Approve**. You will see a pop-up that confirms the new status of the certificate request as Approved. Click **[OK]**.



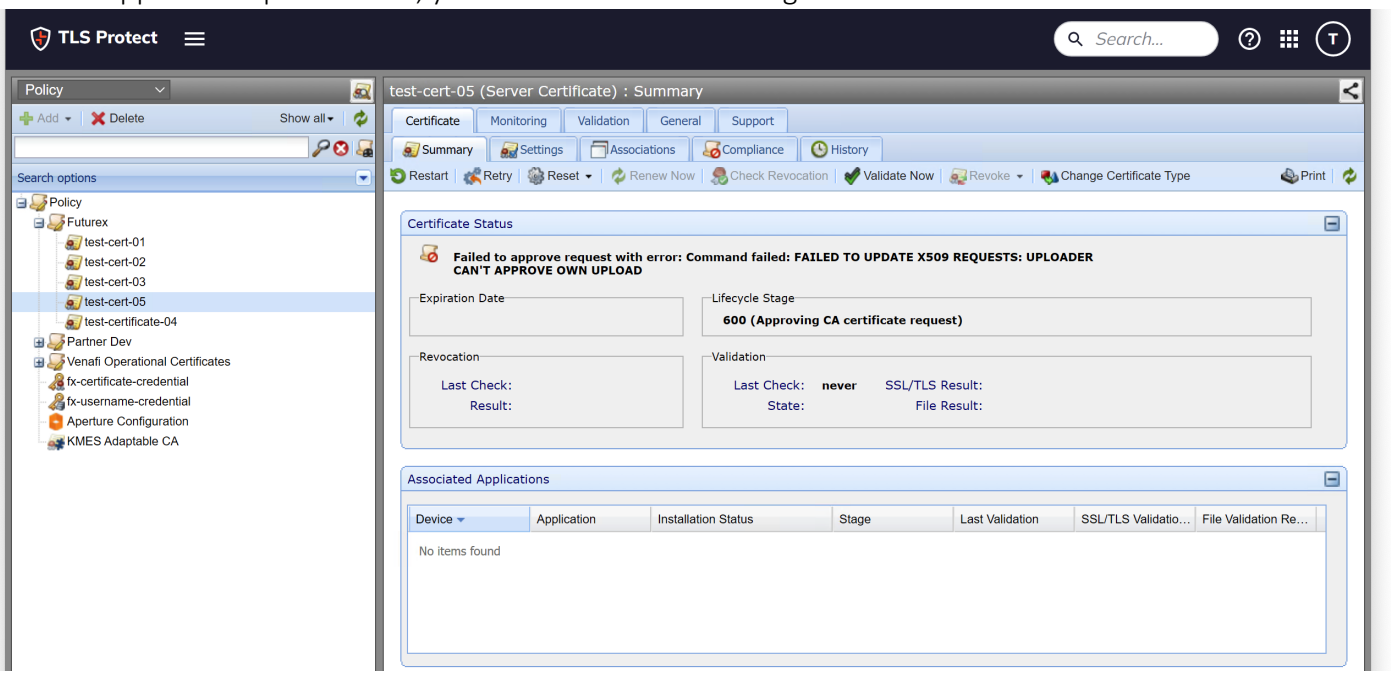
4. In the **Signing Workflow** menu the status of the certificate should show as **Signed**.



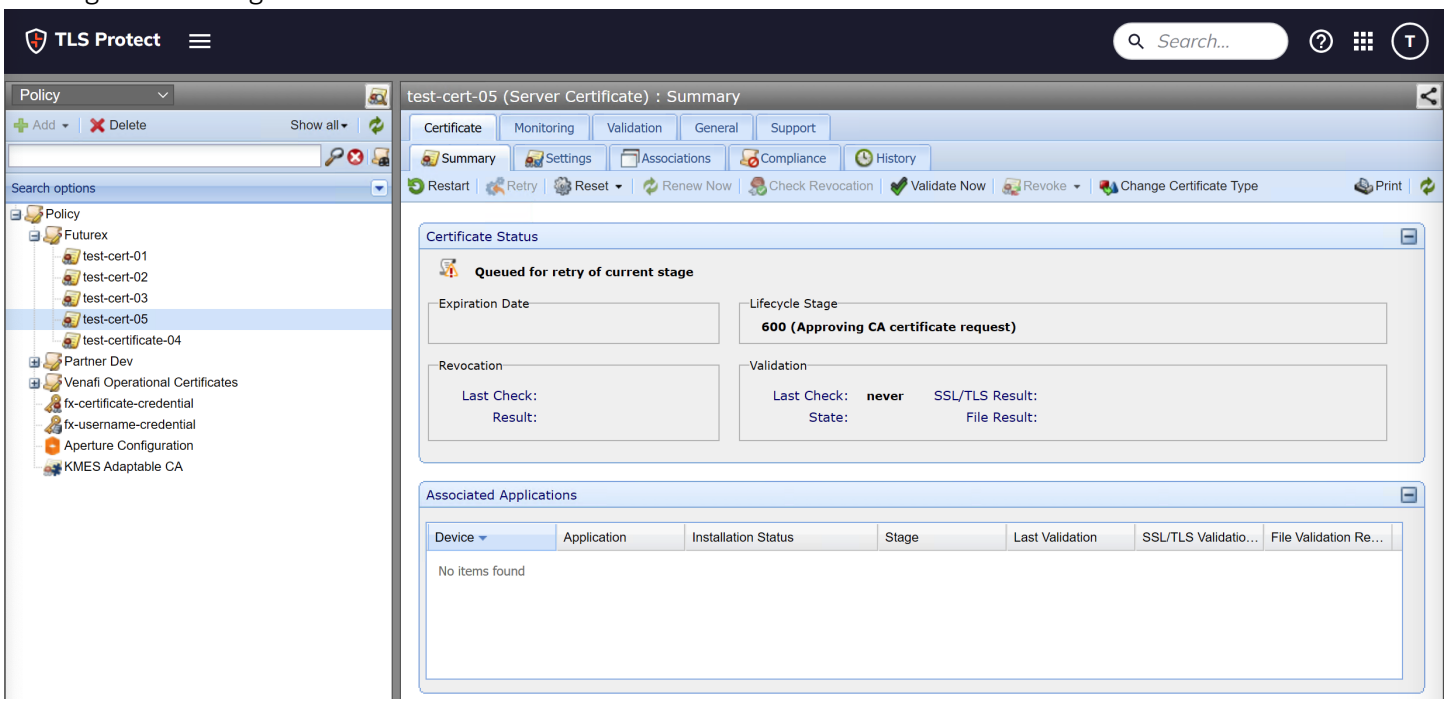
[7.3] VIEW THE ISSUED CERTIFICATE IN VENAFI TPP

To view the KMES-issued certificate in Venafi TPP, perform the following steps:

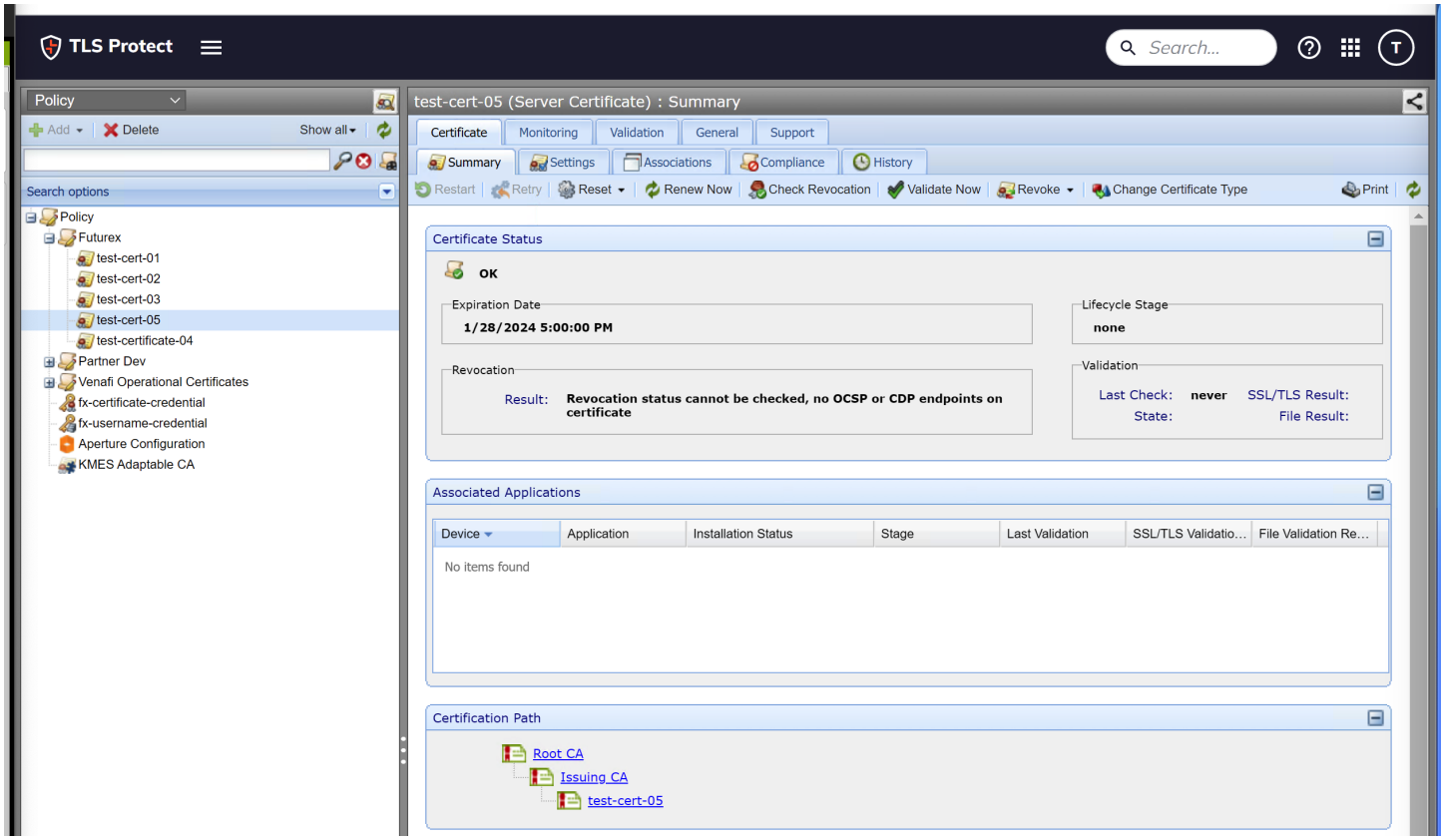
1. Log in to Venafi Trust Protection Platform.
2. Select **Policy Tree** in the main menu.
3. Expand the policy you created in section 6.3 and select the relevant certificate request.
4. If you configured the dual user roles option in section 3.2.2 to separate certificate requesting and certificate approval responsibilities, you should see the following certificate status:



5. Select the **[Retry]** button to retry the request from the current stage. You should see the following message after doing so:



6. Click the refresh button after a few seconds and the certificate status should change to "OK".



[8] TROUBLESHOOTING

[8.1] ERROR MESSAGES

If an error is encountered, the KMES Series 3 will provide an error message that is displayed within Venafi Trust Protection Platform. Most functions provide a description of the error and the entire Excrypt API response containing the error provided by the KMES Series 3's Host API.

When troubleshooting error messages with Futurex's support team, providing the entire message displayed by Trust Protection Platform can speed the diagnosis and resolution.

Error Message	Description
X509 SIGNING NOT ALLOWED	Caused by a lack of upload permissions in the issuance policy.
NO PERMISSIONS	A general error resulting from a lack of permissions for that particular task.
INSUFFICIENT APPROVAL GROUP PERMISSION	A specific error for when a user has view permissions over an approval group but cannot use the approval group.
INVALID REQUEST IDS	Occurs when the correct IDs are used but the approving group doesn't have permissions to view them, so the KMES Series 3 believes them to be invalid.
NOT ALLOWED TO APPROVE YOUR OWN UPLOAD	The same user is being used for both upload and approval. If this is intended, then the restriction can be relaxed in the issuance policy via the "Allow single group for upload and approval" option in the KMES Series 3 configuration.
BAD LOGIN ATTEMPT	User credentials could not be validated. Possible causes include an incorrect or nonexistent username or incorrect permissions.

APPENDIX A: XCEPTIONAL SUPPORT



In today's high-paced environment, we know you are looking for timely and effective resolutions for your mission-critical needs. That is why our Xceptional Support Team does whatever it takes to ensure you have the best experience and support possible. Every time. Guaranteed.

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